Streamlining IT Procurement: Automating Standard Laptop Orders with WI

A Final Year Project Documentation

Project Title	Streamlining IT Procurement: Automating Standard Laptop Orders with WI
Document Type	Project Report / Documentation
Domain	IT Procurement, Workflow Automation, Enterprise Systems
Focus Area	Standard Laptop Ordering Process using Workday Integration (WI)

1. Abstract

This project focuses on improving the way organizations handle the purchase of standard laptops. In many companies, ordering a laptop still involves many manual steps such as filling paper forms, sending emails for approval, checking budgets by hand, and selecting vendors one by one. These steps often cause delays, mistakes, and extra work for the IT and finance teams. The proposed system uses an automated workflow built around Workday Integration (WI) to manage the complete laptop ordering process. It covers raising a procurement request, getting manager approval, performing an automatic budget check, selecting the right vendor, updating inventory, and tracking the order until delivery. The goal is to reduce the time taken for each order, improve accuracy, and give clear visibility to all stakeholders. This documentation explains the problem, the existing manual process, the new automated design, the step-by-step workflow, and the expected benefits of the solution.

2. Introduction

Information Technology (IT) procurement is an important part of any modern organization. Employees need laptops and other devices to do their work. When the process of ordering these devices is slow or confusing, it affects productivity and creates frustration. Many organizations still depend on email chains, shared spreadsheets, and paper-based approvals for standard laptop orders. This traditional approach does not scale well when the company grows or when many employees request devices at the same time.

Workday is a widely used cloud platform for human resources, finance, and procurement activities. Workday Integration (WI) allows different systems and approval steps to talk to each other in an automated way. By connecting the laptop ordering process to WI, organizations can replace manual handoffs with a clear, rule-based workflow. This project studies the current problems in standard laptop procurement and proposes an automated solution that uses WI to manage requests, approvals, budget validation, vendor selection, and inventory updates.

The document is written for final-year engineering students and follows a simple structure so that the ideas can be understood easily. Technical terms related to procurement and workflow automation are explained in everyday language.

3. Problem Statement

The current process for ordering standard laptops has several weaknesses that affect both the requesting employee and the supporting teams:

- **Manual request creation:** Employees fill forms or send emails that may miss important details such as model preference, required accessories, or delivery location.

- **Slow manager approval:** Approval requests sit in email inboxes and can be delayed for days, especially when managers are busy or on leave.
- **No automatic budget check:** Finance or IT teams must open separate systems to verify if enough budget remains in the cost centre. This step is easy to skip or do incorrectly.
- **Inconsistent vendor selection:** Different people choose different vendors without following a standard preferred-vendor list, which can lead to higher prices or longer delivery times.
- **Inventory not updated in real time:** Stock levels of standard laptop models are often maintained in spreadsheets that become outdated quickly.
- **Lack of end-to-end visibility:** Once a request is submitted, the employee and the IT team have limited ways to track its status until the laptop actually arrives.
- **High chance of human error:** Typing mistakes, wrong cost-centre codes, and missed approval steps create rework and compliance issues.

Because of these issues, a simple laptop order that should take a few days can stretch into weeks. The organization also loses money through inefficient processes and missed volume discounts from preferred vendors.

4. Objectives

The main aim of this project is to design and document an automated workflow for standard laptop orders using Workday Integration. The specific objectives are:

1. To study the existing manual laptop procurement process and identify the main pain points.
2. To design a clear automated workflow that covers request submission, manager approval, budget check, vendor selection, and inventory update.
3. To define how Workday Integration (WI) can connect the different steps so that data moves without manual re-entry.
4. To reduce the average time required to process a standard laptop order.
5. To improve data accuracy by applying validation rules at each stage of the workflow.
6. To give employees and managers a simple way to track the status of their requests.
7. To create documentation that can be used as a reference for implementation or further student projects.

5. Existing System

In the existing system, the laptop ordering process is mostly manual and depends on people remembering the correct sequence of steps. A typical flow looks like this:

- **Request initiation:** The employee downloads a request form or writes an email to the IT helpdesk describing the need for a laptop.
- **Manager approval via email:** The IT team or the employee forwards the request to the reporting manager. The manager replies with approval or asks for more information.
- **Manual budget verification:** After approval, someone from finance or IT logs into the finance system or opens a budget spreadsheet to confirm that funds are available under the correct cost centre.
- **Vendor selection:** The procurement team or IT asset team chooses a vendor based on personal knowledge or old price lists. There is often no forced use of preferred vendors.
- **Purchase order creation:** A purchase order is created in a separate procurement tool or even in a spreadsheet and then emailed to the vendor.
- **Inventory update:** When the laptop arrives, the asset register or inventory sheet is updated by hand. Sometimes this step is forgotten.
- **Delivery and confirmation:** The laptop is handed over to the employee and a confirmation email may or may not be sent.

This process works for a small number of requests, but it becomes difficult to manage when dozens of orders arrive every month. There is no single system of record, and status tracking depends on searching through email threads.

6. Proposed System

The proposed system replaces the scattered manual steps with one connected automated workflow powered by Workday Integration (WI). All key actions happen inside or through Workday, so that data does not need to be copied from one tool to another.

Key features of the proposed system include:

- **Self-service procurement request:** Employees raise a standard laptop request from a simple form inside Workday or a connected portal. The form already contains preferred models and required fields.
- **Automated manager approval:** The system routes the request to the correct manager based on the organization hierarchy. Reminder notifications are sent if the approval is delayed.
- **Real-time budget check:** After manager approval, the workflow automatically checks the available budget for the selected cost centre. If the budget is insufficient, the request is returned with a clear message.
- **Rule-based vendor selection:** The system picks a preferred vendor according to pre-defined rules (price, delivery time, contract status). Manual override is possible only for special cases.
- **Automatic inventory update:** When a laptop is ordered or received, the inventory record is updated so that stock levels always stay current.
- **End-to-end status tracking:** Both the employee and the IT team can see the current stage of the request (Submitted, Approved, Budget Checked, Ordered, Delivered) in one place.
- **Audit-ready logs:** Every action is recorded with time stamps and user details, which helps in audits and process improvement.

By using WI, the system can also send and receive data from other enterprise tools such as the asset management system or the vendor portal without requiring users to log into multiple applications.

7. System Architecture & Workflow

The architecture is designed as a linear yet flexible workflow. Each step is triggered automatically when the previous step is completed successfully. The main components are the employee self-service portal, Workday as the central orchestration engine, the approval engine, the budget service, the vendor selection rules, and the inventory module.

Below is the step-by-step laptop order automation workflow:

Step 1: Raise Procurement Request

The employee logs into the self-service portal linked to Workday and selects the standard laptop catalogue. They fill in the required details (model, quantity, delivery location, cost centre, and business justification). On submission, a unique request ID is generated and the request enters the workflow.

Step 2: Manager Approval

Workday routes the request to the employee’s reporting manager. The manager receives a notification and can approve, reject, or ask for more information. If approved, the request moves forward. If rejected, the employee is notified with the reason.

Step 3: Automatic Budget Check

The system calls the budget service (via WI) and verifies that the selected cost centre has enough remaining budget for the laptop cost. If the budget is available, the request continues. If not, the request is returned to the employee or finance for review.

Step 4: Vendor Selection

Based on configured rules (preferred vendor list, contract rates, delivery SLA), the system selects the most suitable vendor for the chosen laptop model. The selected vendor and estimated price are recorded against the request.

Step 5: Purchase Order & Inventory Reservation

A purchase order is created automatically and sent to the selected vendor. At the same time, the inventory module reserves or updates the expected stock quantity so that other teams can see the pending order.

Step 6: Order Tracking & Delivery Confirmation

The workflow waits for the vendor confirmation and shipping updates. When the laptop is received and checked by the IT asset team, the inventory is updated to “in stock / assigned” and the employee is notified that the device is ready for collection or delivery.

Step 7: Closure & Reporting

The request is marked as completed. All timestamps, approvals, and cost data are stored for reporting and audit purposes. Management dashboards can show average cycle time, budget utilisation, and vendor performance.

This step-by-step design keeps the process transparent and reduces the need for people to chase each other for status updates. Exceptions (for example, custom configurations or budget shortages) are handled through defined exception paths so that the main flow stays clean.

8. Benefits

Implementing the automated laptop ordering workflow brings several practical advantages:

- **Faster turnaround:** Standard orders that previously took one to two weeks can often be completed in a few days because waiting time between steps is removed.
- **Fewer errors:** Validation rules catch missing fields, wrong cost centres, and insufficient budgets before the order is placed.
- **Better budget control:** Real-time budget checks prevent overspending and give finance teams clearer visibility of committed costs.
- **Consistent vendor usage:** Preferred vendors are selected by rules, which helps the organisation get better prices and reliable delivery.
- **Accurate inventory:** Stock levels stay up to date, reducing the chance of ordering models that are already available or out of stock.
- **Improved employee experience:** Employees can track their requests themselves instead of sending repeated follow-up emails.
- **Stronger compliance and audit trail:** Every action is logged, making it easier to answer audit questions and improve the process over time.
- **Scalability:** The same workflow can handle a higher volume of requests without adding proportional manual effort.

9. Conclusion

Manual processes for ordering standard laptops create delays, errors, and unnecessary work for employees, managers, and support teams. This project has examined those problems and proposed a practical automated solution built around Workday Integration. By connecting procurement request submission, manager approval, budget checking, vendor selection, and inventory updates into one clear workflow, organisations can process laptop orders faster and with greater accuracy.

The design keeps the language and steps simple so that it can be understood and implemented by student teams or by IT departments looking for a starting point. While the current focus is on standard laptop models, the same pattern can be extended to other categories of IT equipment. Overall, the automated approach offers a realistic way to streamline IT procurement and free up people's time for higher-value work.

10. Future Scope

The current design covers the core laptop ordering flow. Several useful extensions can be added later:

- Support for non-standard or custom laptop configurations with additional approval levels.
- Integration with asset tagging and employee onboarding processes so that a new joiner's laptop is ordered automatically.
- Mobile-friendly request and approval screens for managers who travel frequently.
- Predictive analytics to forecast laptop demand based on hiring plans and replacement cycles.
- Direct punch-out catalogues from preferred vendors so that real-time pricing and availability are shown inside the request form.
- Extension of the same workflow pattern to monitors, docking stations, and other standard accessories.
- Chatbot or virtual assistant support that guides employees while they raise a request.

These enhancements can be taken up as follow-up projects or as continuous improvement work after the basic automation is live.

11. References

[1] Workday Documentation – Procurement and Business Process Framework (official product guides).

[2] ITIL Foundation materials on Service Request Management and Asset Management practices.

[3] General literature on procure-to-pay (P2P) process automation and workflow design.

[4] Academic papers and case studies on enterprise resource planning (ERP) integration for purchasing.

[5] Internal organisational procurement policies and preferred vendor guidelines (where applicable).

[6] Student project reports and open educational resources on business process automation.